

Making shapes

As well as adding windows, buttons, and colors to a graphical user interface (GUI), Tkinter can also be used to draw shapes.

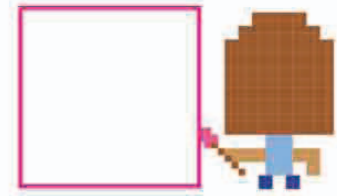
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Creating basic shapes

Rectangles and ovals are useful shapes for drawing all sorts of things. Once a canvas has been created, the following functions can be used to draw shapes on it.



```
>>> from tkinter import *
>>> window = Tk()
>>> drawing = Canvas(window, height=500, width=500)
>>> drawing.pack()
>>> rect1 = drawing.create_rectangle(100, 100, 300, 200)
>>> square1 = drawing.create_rectangle(30, 30, 80, 80)
>>> oval1 = drawing.create_oval(100, 100, 300, 200)
>>> circle1 = drawing.create_oval(30, 30, 80, 80)
```

Creates a canvas to draw on

Sets the size of the canvas

Sets the position and size of the rectangle using coordinates (see below)

Draws a rectangle

A square can be made by drawing a rectangle with all sides the same length

Draws a circle

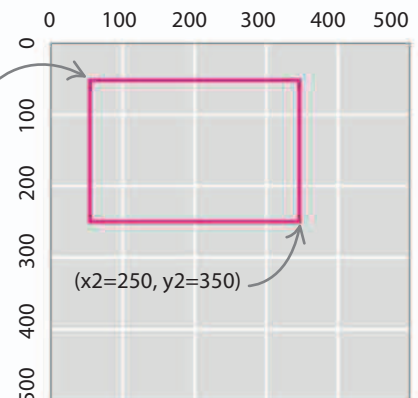
Sets the position and size of the circle

Drawing with coordinates

Coordinates are used to tell the computer exactly where to create shapes. The first number ("x") tells the computer how far along the screen to go. The second number ("y") tells the computer how far down to go.

▽ Coordinates grid

The top-left corner of the rectangle is at coordinates (50, 50). The bottom-right corner is at (250, 350).



```
>>> drawing.create_rectangle(50, 50, 250, 350)
```

This is the name of the canvas

Coordinates for the top left of the rectangle

Coordinates for the bottom right of the rectangle

△ Setting the coordinates

The first two numbers give the coordinates for the top-left corner of the rectangle. The second two numbers locate the bottom-right corner.